



Apache Airflow Installation and Validation Guide

Overview

This document provides a **complete, step-by-step guide** to installing and validating **Apache Airflow** on a local machine. Two supported installation methods are covered:

1. **Local installation using Python virtual environment**
2. **Docker-based installation using Docker Compose**

You may choose either approach based on your system setup and learning goals. Both methods result in a fully functional Airflow environment with a web-based UI for managing workflows.

Prerequisites

Before starting, ensure the following requirements are met.

Common Requirements

- Stable internet connection
- Basic familiarity with using the terminal or command prompt

For Local Installation

- Python **3.8 to 3.11**
- Python added to system PATH

For Docker Installation

- Docker Desktop installed
- Docker daemon running



- Docker Compose enabled

Note: Installation steps may vary slightly depending on your operating system (Windows, macOS, or Linux) and system permissions.

Choosing an Installation Method

Method	Recommended When
Local (Python virtual environment)	You want to understand Airflow dependencies and work locally with Python
Docker (Docker Compose)	You want a production-like setup with minimal local dependency management

Method 1: Local Installation Using Python Virtual Environment

Step 1: Verify Python Version

Check the installed Python version:

```
python --version
```

Ensure the version is between **Python 3.8 and 3.11**.

Step 2: Create a Virtual Environment

Create a virtual environment to isolate Airflow dependencies:

```
python -m venv airflow_env
```



Step 3: Activate the Virtual Environment

- **Windows**

```
airflow_env\Scripts\activate
```

- **macOS / Linux**

```
source airflow_env/bin/activate
```

Once activated, all Python packages will be installed inside this environment.

Step 4: Install Apache Airflow

Apache Airflow must be installed using a **constraints file** to ensure compatible dependency versions.

```
pip install "apache-airflow==<AIRFLOW_VERSION>" --constraint "<CONSTRAINTS_URL>"
```

Replace:

- <AIRFLOW_VERSION> with the desired Airflow version
- <CONSTRAINTS_URL> with the corresponding constraints file URL from the official Airflow documentation

Step 5: Verify Installation

Confirm that Airflow is installed correctly:

```
airflow version
```

This command should return the installed Airflow version.

Step 6: Initialise the Metadata Database

Initialise Airflow's metadata database:



```
airflow db init
```

This sets up internal tables required for DAGs, task runs, and scheduling.

Step 7: Create an Admin User

Create a user to log in to the Airflow UI:

```
airflow users create
```

You will be prompted to enter:

- Username
- Password
- Role (use Admin)
- Email

Step 8: Start Airflow Services

Open two terminals (ensure the virtual environment is active in both).

Terminal 1 – Webserver

```
airflow webserver --port 8080
```

Terminal 2 – Scheduler

```
airflow scheduler
```

Method 2: Docker Installation Using Docker Compose

Step 1: Verify Docker Installation

Check Docker and Docker Compose versions:



```
docker --version  
docker compose version
```

Ensure Docker is running before proceeding.

Step 2: Create a Project Directory

Create and move into a new directory for Airflow:

```
mkdir airflow-docker  
cd airflow-docker
```

Step 3: Download the Official Docker Compose File

Download the official `docker-compose.yaml` file from the Apache Airflow documentation and place it in this directory.

Using the official file ensures correct service configuration and compatibility.

Step 4: Configure Environment Variables

Create a `.env` file in the same directory and add:

```
AIRFLOW_UID=50000
```

This helps prevent file permission issues, especially Linux systems.

Step 5: Initialise Airflow

Run the following command:

```
docker compose up airflow-init
```



This step:

- Initialises the metadata database
- Creates default users
- Prepares required services

Step 6: Start Airflow Services

Start all Airflow services:

```
docker compose up
```

This will start the following containers:

- Webserver
- Scheduler
- Database (Postgres)
- Redis
- Triggerer

Validation and Accessing the Airflow UI

Once Airflow is running (using either method), open a browser and navigate to:

<http://localhost:8080>

Validate the Installation

Confirm the following:

- The Airflow login page loads
- You can log in successfully
- The Airflow home page opens
- Example DAGs are visible
- You can trigger an example DAG and observe task execution



Successful completion of these steps confirms that Airflow is installed correctly.

Stopping Airflow

Local Installation

Press `Ctrl + C` in both the webserver and scheduler terminals.

Docker Installation

Run:

```
docker compose down
```

Common Troubleshooting

Issue	Possible Cause
Airflow fails to start	Unsupported Python version
Port 8080 unavailable	Another service using the port
Dependency conflicts	Missing or incorrect constraints file
Docker containers exit	Docker daemon not running
Permission errors	Incorrect AIRFLOW_UID value

Next Steps

After successful installation, you are ready to:

- Create and modify DAGs
- Schedule workflows
- Monitor task execution
- Explore retries, logs, and dependencies



This installation serves as the foundation for all future Apache Airflow usage.

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